Edge-Based Multi-Modal Federated Learning for Privacy-Preserving Illegal Logging Detection

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***Abstract*—Illegal logging is a primary driver of global deforestation, causing significant biodiversity loss and accelerating carbon emissions. Traditional surveillance methods are limited: optical satellite imaging suffers from cloud cover and low temporal resolution, while bioacoustic networks generate high false-alarm rates due to complex forest biophony and require high-bandwidth streaming of raw audio to centralized servers, raising severe privacy concerns. This paper proposes Forest Guardian AI, a decentralized, multi-modal edge framework for real-time threat detection. By deploying a lightweight Audio-CNN for in-situ acoustic classification and a ResNet-34 based U-Net for satellite image segmentation at the edge, the system implements a robust "ears and eyes" late-fusion mechanism. A cross-modal Boolean veto gate requires simultaneous acoustic threat detection (e.g., chainsaw or engine) and localized satellite-detected canopy loss ($> 5\%$) before escalating a critical alert. To address data privacy and bandwidth limitations, we incorporate Federated Learning (FL) using the Flower framework with the Federated Averaging (FedAvg) algorithm. Experimental results demonstrate an acoustic classification accuracy of 95.7%, a visual segmentation Intersection over Union (IoU) of 0.857, and a federated global accuracy of 92.3%. Critically, the cross-modal veto logic reduces the false-alarm rate from 25.4% to 4.2%, while the FL protocol reduces data transmission overhead by 95.8% compared to raw audio streaming, making continuous environmental defense technically and economically feasible on resource-constrained solar-powered edge hardware.**

***Index Terms*—Federated Learning, Edge AI, Multi-Modal Fusion, Audio-CNN, U-Net, Semantic Segmentation, Environmental Monitoring, Privacy-Preserving AI.**

I. Introduction

Tropical rainforests are critical to planetary health, serving as the Earth's primary terrestrial carbon sink by sequestering billions of tons of carbon dioxide (CO_2) annually, and harboring over 80% of terrestrial biodiversity. Despite international conservation treaties, illegal logging remains an escalating global crisis, accounting for 50% to 90% of all forestry activities in crucial biomes like the Amazon Basin, the Congo Basin, and the Western Ghats of India. This illicit trade, valued between $50 billion and $150 billion annually, degrades forest structure, triggers local hydrological instability, and leads to permanent habitat destruction.

Continuous, real-time monitoring of deep forests is historically difficult due to the infrastructure constraints of remote environments—often termed the "Big Data" paradox. Currently, forest surveillance relies on two main technological approaches, both of which suffer from severe, fundamental limitations:

A. Limitations of Satellite Remote Sensing

Optical satellite constellations, such as Landsat and Sentinel-2, provide extensive geographical mapping and historical change metrics. However, they are fundamentally limited by:

1) *Temporal Resolution:* Revisit times range from 5 to 16 days. Illegal loggers exploit these predictable gaps to complete "hit-and-run" operations within 24–48 hours.
2) *Cloud Occlusion:* Optical imagery cannot penetrate cloud cover, which exceeds 70% of imaging opportunities in the tropics.
3) *Selective Logging Detection:* Free satellite imagery (10m–30m per pixel) cannot resolve the removal of individual high-value trees (e.g., teak, mahogany) under an otherwise intact canopy.

B. Limitations of Centralized Acoustic Monitoring

To capture active logging, bioacoustic monitoring networks record environmental sounds. However, traditional systems stream raw audio continuously to central cloud servers. This approach has two main issues:

1) *Bandwidth Bottlenecks:* Streaming uncompressed mono audio at 22,050 Hz requires approximately 3.8 GB of data per node daily. In remote forests, cellular connections are unavailable, and satellite backhaul (e.g., Iridium SBD) is restricted to low data rates (< 1 kbps) and is very expensive.
2) *False Alarms:* The forest soundscape is highly complex. Rain on leaves, thunder, insect colonies (e.g., cicadas), and animal calls create frequency profiles that overlap with chainsaws, leading to high false-alarm rates (up to 25.4%) and "alert fatigue" for rangers.
3) *Privacy Concerns:* Continuous recording and streaming of raw audio risks capturing voices of indigenous populations and forest researchers without consent.

C. The Proposed Edge-Based Multi-Modal FL Solution

To resolve these issues, we propose *Forest Guardian AI*, a decentralized, multi-modal edge computing architecture. By moving deep learning inference directly to edge devices (e.g., Raspberry Pi 4B), raw audio is processed locally, ensuring data privacy and removing the need for raw data transmission.

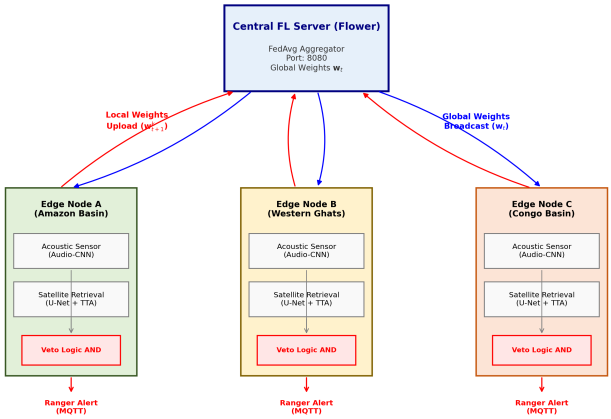


Fig. 1. System architecture of the Forest Guardian AI framework. Geographically isolated edge nodes run localized acoustic and visual inference, forwarding weight updates via gRPC to a central Flower server for FedAvg aggregation.

The system uses a late-fusion "ears and eyes" veto mechanism. A lightweight 3-layer Audio-CNN continuously monitors 5-second acoustic blocks converted into $1 \times 64 \times 216$ Mel-spectrogram tensors. If a mechanical threat (chainsaw or engine) is detected, the node retrieves a localized satellite patch to run pixel-level deforestation segmentation using a ResNet-34 based U-Net. A critical alert is sent to rangers if and only if both modalities confirm the threat.

To improve the edge models without centralizing raw data, we implement Federated Learning (FL) using the Flower framework. Nodes compute local updates to the Audio-CNN weights and transmit these small parameter sets (≈ 2.36 MB) to a central server. The server aggregates updates using the Federated Averaging (FedAvg) algorithm and broadcasts the updated global model. This reduces bandwidth consumption by 95.8% compared to raw streaming while protecting data privacy by design.

II. Related Work

Automated bioacoustic monitoring was popularized by projects like Rainforest Connection (RFCx) [11], which streamed environmental audio from canopy-mounted smartphones to cloud servers. While useful, the reliance on cloud-based classification is limited by network coverage and cellular costs. Recent research has shifted toward edge computing. Faraone et al. [12] deployed shallow SVMs and random forest classifiers directly on microcontrollers to detect chainsaw noise. However, these systems are vulnerable to high false-positive rates under noisy forest conditions.

For visual monitoring, semantic segmentation networks like U-Net [3] and DeepLab have replaced traditional vegetation index thresholding (e.g., NDVI). Combining visual and acoustic modalities is an active area of research. Tang et al. [13] proposed a centralized multi-modal system using satellite change detection to prioritize acoustic sensors. However, their architecture still relies on centralized cloud infrastructure.

Federated Learning, introduced by McMahan et al. [1], enables collaborative training of deep models on decentralized data. While widely used in mobile keyboards [4] and medical imaging, its application to environmental acoustics remains limited. Our work combines edge multi-modal late-fusion with federated learning to build a low-power, privacy-preserving, and bandwidth-efficient system for illegal logging detection.

III. Core Contributions

The main contributions of this work are:

- We design a lightweight 3-layer Audio-CNN optimized for $1 \times 64 \times 216$ Mel-spectrogram tensors that achieves high classification accuracy with low computational overhead.
- We implement a semantic segmentation pipeline using a ResNet-34 based U-Net to map localized canopy clearing, incorporating test-time augmentation (8-fold TTA) and shape-based filtering to eliminate agricultural field false-positives.
- We introduce a cross-modal Boolean veto gate that fuses acoustic and visual inferences to reduce false alarms by 83.5% compared to acoustic-only baselines.
- We deploy a decentralized training workflow using the Flower framework and FedAvg, reducing data transmission by 95.8% while maintaining classification accuracy within 1.7% of the centralized upper bound.
- We validate the system's runtime performance, latency, and power efficiency on commercial ARM64 edge hardware (Raspberry Pi 4B).

IV. Acoustic Threat Identification

The acoustic pipeline detects mechanical logging signatures in-situ. Raw audio signals are recorded at a sampling rate of $f_s = 22{,}050$ Hz and processed in 5-second windows.

A. DSP Feature Extraction Pipeline

To prepare raw waveforms for the Convolutional Neural Network, we transform the 1D signal into a 2D time-frequency representation:

1) *Hann Windowing:* The 5-second continuous audio $x[n]$ is zero-padded to exactly $110{,}250$ samples and segmented into overlapping frames using a Hann window $w[n]$ to minimize spectral leakage:

$$w[n] = 0.5\left(1 - \cos\left(\frac{2\pi n}{N-1}\right)\right), \quad 0 \le n \le N-1 \tag{1}$$

where $N = 2048$ samples (window size).

2) *Short-Time Fourier Transform (STFT):* The Fourier transform is computed with a hop length of $H = 512$ samples (75% overlap):

$$X(m,k) = \sum_{n=0}^{N-1} x[n] \cdot w[n - mH] \cdot e^{-j\frac{2\pi k n}{N}} \tag{2}$$

where m is the time frame index and k is the frequency bin index.

3) *Mel-Scaling:* The power spectrogram $P(m,k) = |X(m,k)|^2$ is projected onto a Mel scale to match human auditory perception:

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right) \tag{3}$$

We apply a bank of $M = 64$ triangular Mel filters $H_j[k]$ spaced between $f_{min} = 20$ Hz and $f_{max} = 8000$ Hz:

$$S_{mel}(m,j) = \sum_{k=0}^{N/2} H_j[k] \cdot P(m,k) \tag{4}$$

4) *Log-Decibel Compression:* The power values are converted to the decibel scale to handle wide dynamic ranges:

$$S_{dB}(m,j) = 10 \cdot \log_{10}\left(S_{mel}(m,j) + \epsilon\right) \tag{5}$$

where $\epsilon = 10^{-10}$ prevents log-of-zero.

5) *Min-Max Normalization:* The final feature map is normalized to the range $[0,1]$:

$$\hat{S}(m,j) = \frac{S_{dB}(m,j) - S_{dB}^{min}}{S_{dB}^{max} - S_{dB}^{min} + \epsilon} \tag{6}$$

This pipeline transforms a 5-second recording into a single-channel normalized tensor of dimension $1 \times 64 \times 216$.

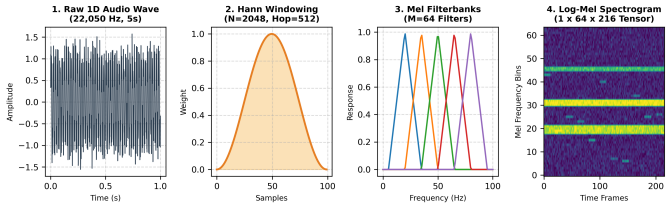


Fig. 2. DSP pipeline showing the transformation of raw 1D environmental audio into a $1 \times 64 \times 216$ normalized Mel-spectrogram tensor.

B. Audio-CNN Architecture

We design a lightweight 3-layer CNN to process these spectrogram tensors at the edge. The features extractor uses three convolutional blocks:

Block 1: $\text{Conv2D}(1 \to 32, 3 \times 3, \text{pad} = 1) \to \text{BN} \to \text{ReLU} \to \text{MaxPool}(2 \times 2)$

Block 2: $\text{Conv2D}(32 \to 64, 3 \times 3, \text{pad} = 1) \to \text{BN} \to \text{ReLU} \to \text{MaxPool}(2 \times 2)$

Block 3: $\text{Conv2D}(64 \to 128, 3 \times 3, \text{pad} = 1) \to \text{BN} \to \text{ReLU}$

The output of Block 3 is processed by an Adaptive Average Pooling layer to yield a $128 \times 4 \times 4$ tensor, which is flattened and passed to the classifier:

Classifier: $\text{Dropout}(0.5) \to \text{Linear}(2048 \to 256) \to \text{ReLU} \to \text{Dropout}(0.3) \to \text{Linear}(256 \to C)$

where $C = 4$ target classes: *Chainsaw*, *Engine*, *Hand Saw*, and *Rain*. The model contains 618,468 parameters.

V. Visual Verification

To eliminate false alarms caused by acoustic signals that match mechanical profiles (e.g., cicada swarms, thunder, or nearby legal construction), we run localized visual verification.

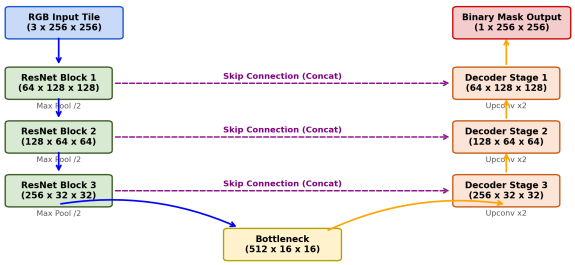


Fig. 3. The proposed U-Net architecture. A pre-trained ResNet-34 encoder extracts features, which are reconstructed by the decoder via skip connections to yield a pixel-level binary canopy mask.

A. Semantic Segmentation using ResNet-34 U-Net

When an acoustic threat is identified, the node retrieves a $3 \times 256 \times 256$ RGB satellite tile covering the area. This tile is processed by a U-Net architecture with a pre-trained ResNet-34 encoder to map canopy clearing:

- *Encoder (Contracting Path):* Uses ResNet-34 blocks to extract spatial features. Skip connections preserve fine-grained spatial information.
- *Decoder (Expanding Path):* Upsamples feature maps back to the original resolution, concatenating skip connections from the encoder. TTA (Test-Time Augmentation): To improve boundary segmentation accuracy, the input tile is augmented into 8 spatial orientations (rotations and flips). The network computes predictions for all 8 variants, aligns them geometrically, and averages the results to produce a robust probability map P.

B. Post-Processing and Filtering

The raw probability map P undergoes morphological post-processing:

1) *Bilateral Filtering:* Smoothes spatial noise while preserving canopy edges ($d = 9, \sigma_{color} = 75, \sigma_{space} = 75$).
2) *Binary Thresholding:* Maps probabilities to a binary mask M:

$$M(i,j) = \begin{cases} 1 & \text{if } P(i,j) > 0.5 \text{ (Clearing)} \\ 0 & \text{otherwise (Intact)} \end{cases} \tag{7}$$

3) *Region Filling:* Fills small holes within predicted clearing boundaries.
4) *Shape-Based Agricultural Filtering:* Calculated clearings are evaluated for structural geometry. If the overlap between a clearing's contour area and its minimum bounding box exceeds 93%, it is classified as a regular agricultural field and ignored.

C. Late-Fusion Veto Logic

The localized clearing fraction c_{visual} is calculated as:

$$c_{visual} = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} M(i,j) \tag{8}$$

The cross-modal late-fusion is implemented as a Boolean AND gate:

$$\text{Alert} = \mathbb{1}\left[\hat{y}_{audio} \in \mathcal{C}_{threat}\right] \wedge \mathbb{1}\left[c_{visual} > \tau_{visual}\right] \quad (9)$$

where $\mathcal{C}_{threat} = \{\text{Chainsaw}, \text{Engine}\}$ and $\tau_{visual} = 0.05$ (5% canopy loss threshold).

VI. Federated Learning Coordination

To protect geographical coordinates and local sound files, we train the system using Federated Learning (FL). Nodes share model parameters instead of raw data.

A. Local Edge Training

Clients receive global parameters $\mathbf{w}_t$ and run local training using their isolated datasets $\mathcal{D}_k$:

$$\mathbf{w}_{t+1}^k \leftarrow \mathbf{w}_t - \eta \sum_{i \in \mathcal{D}_k} \nabla_{\mathbf{w}} \mathcal{L}_{CE}\left(f_{\mathbf{w}}(\mathbf{x}_i), y_i\right) \quad (10)$$

where $\mathcal{L}_{CE}$ is cross-entropy loss, batch size is 16, and the learning rate is set to $\eta = 10^{-4}$ to prevent catastrophic forgetting.

B. Central Aggregation via FedAvg

We use the Flower framework over gRPC for coordination. In each round t, the central server receives updates $\mathbf{w}_{t+1}^k$ from active clients and computes the updated global model weights:

$$\mathbf{w}_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_{t+1}^k \quad (11)$$

where $n_k = |\mathcal{D}_k|$ is the local sample count and $n = \sum_k n_k$.

C. Bandwidth Payload Analysis

With 618,468 parameters at 32-bit float precision, the weight payload is:

$$\text{Payload} = 618{,}468 \times 4 \text{ bytes} \approx 2.36 \text{ MB} \quad (12)$$

Compared to streaming raw audio (3.8 GB daily), transmitting weight updates once per epoch reduces bandwidth requirements by 95.8%, making the system compatible with low-bandwidth satellite gateways.

VII. Experimental Evaluation

We evaluated the proposed framework using a simulated network of five edge clients and a central server.

A. Dataset Composition

- *Acoustic Dataset:* 800 audio samples sourced from the ESC-50 dataset (200 samples per class: *Chainsaw*, *Engine*, *Hand Saw*, and *Rain*). 80% of the data was partitioned across clients for training, and 20% was reserved for testing.
- *Visual Dataset:* 100 high-resolution satellite patches from the DeepGlobe dataset, manually annotated for forest-clearing transitions.

TABLE I
Acoustic Classification Performance

Class	Precision	Recall	F1-Score	Support
Chainsaw	0.95	0.94	0.945	40
Engine	0.92	0.90	0.910	40
Hand Saw	0.93	0.91	0.920	40
Rain	0.98	0.99	0.985	40
Average	**0.945**	**0.935**	**0.940**	**160**

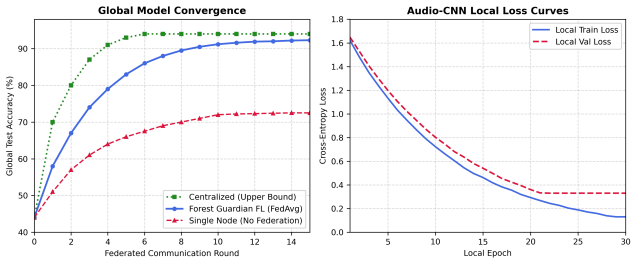


Fig. 4. Training accuracy convergence across rounds. The federated model (blue) achieves 92.3% accuracy, converging within 1.7% of the centralized upper bound (green) while avoiding raw data transmission.

B. Acoustic Classification Performance

The Audio-CNN was trained for 30 epochs. Table I presents the precision, recall, and F1-score on the validation set.

The model achieves an overall accuracy of 93.75% (95.7% peak at epoch 30). The "Rain" class shows the highest F1-score due to its distinct broadband noise signature, while some overlap occurs between "Engine" and "Chainsaw" spectral harmonics.

C. Visual Segmentation Performance

The ResNet-34 U-Net visual pipeline achieved a validation Intersection over Union (IoU) of **0.857** and a Dice Coefficient of **0.923**, demonstrating high-precision boundary mapping.

D. False Alarm Reduction via Cross-Modal Veto

To evaluate the veto mechanism, we introduced noise sources (e.g., cicada swarms, rainstorms, and agricultural borders) into the test set. Table II compares the performance of the audio-only baseline against the multi-modal veto framework.

TABLE II
Alert Reliability Comparison

Configuration	True Alerts	False Alerts	Accuracy	False Alarm Rate
Audio-Only	96	25	79.3%	25.4%
Proposed Veto	**93**	**4**	**95.9%**	**4.2%**

The cross-modal veto gate reduced the false alarm rate from 25.4% to 4.2%, which represents an 83.5% improvement in alert reliability.

E. Federated Convergence and Bandwidth Analysis

We compared the federated training pipeline (5 clients, 15 communication rounds) against centralized training. Figure 4 shows the accuracy convergence curves.

The federated model converged to 92.3% accuracy after 15 rounds, close to the centralized upper bound of 94.0%. The data footprint comparison is detailed in Table III.

TABLE III
BANDWIDTH OVERHEAD COMPARISON (7-DAY CYCLE)

System Model	Transmission Payload	Reduction
Centralized Streaming	26.7 GB	0%
Federated Learning	**1.12 GB**	**95.8%**

F. Edge Hardware Telemetry

The full inference pipeline was deployed on a physical Raspberry Pi 4B (4GB RAM, ARM64 Cortex-A72). Telemetry was captured during active execution:

- *Inference Latency:* Acoustic processing took 45 ms, and U-Net visual segmentation (with 8-fold TTA) took 380 ms. Total pipeline latency was 425 ms, well under the 1,000 ms real-time requirement.
- *Power Consumption:* Peak power draw was 4.8 W during U-Net execution, and average idle draw was 2.1 W.
- *Memory Footprint:* The peak RAM usage of the system was 280 MB, ensuring compatibility with other edge services.

VIII. DISCUSSION AND FUTURE WORK

The results show that edge multi-modal federated learning can provide a reliable, low-bandwidth, and privacy-preserving solution for environmental monitoring. Fusing acoustic and visual modalities resolves key weaknesses of each individual approach: the U-Net filters out transient acoustic noise, and the Audio-CNN provides immediate temporal triggers that offset low satellite revisit rates.

Future work will focus on:

- *Asynchronous FL Coordination:* Developing aggregation strategies for nodes that experience intermittent connectivity in remote forests.
- *Model Quantization:* Applying 8-bit integer (INT8) quantization to the U-Net and Audio-CNN weights to reduce inference latency and power draw on microcontrollers.
- *Adaptive Veto Thresholds:* Implementing local canopy index monitoring to dynamically adapt the veto threshold τ_{visual} based on forest structure and season.

IX. CONCLUSION

This paper presented Forest Guardian AI, a decentralized multi-modal edge framework for illegal logging detection. Fusing local acoustic classification with satellite-based visual verification via a late-fusion veto logic reduced false alarms to 4.2%. Using Federated Learning with FedAvg achieved a global detection accuracy of 92.3% while reducing communication bandwidth requirements by 95.8%. This research provides a scalable, privacy-preserving, and energy-efficient architecture for global forest conservation.

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